BCD to 7-Segment Display — Java

Java equivalent using a switch statement. Segment order is a, b, c, d, e, f, g; 1 = ON and 0 = OFF.

```
public class BCDTo7Segment {
    public static void main(String[] args) {
        int bcd = 7;

        int a = (bcd >> 3) & 1;
        int b = (bcd >> 2) & 1;
        int c = (bcd >> 1) & 1;
        int d = bcd & 1;

        int[] seg = new int[7];

        switch (bcd) {
            case 0: seg = new int[]{1, 1, 1, 1, 1, 1, 0}; break;
            case 1: seg = new int[]{0, 1, 1, 0, 0, 0, 0}; break;
            case 2: seg = new int[]{1, 1, 0, 1, 1, 0, 1}; break;
            case 3: seg = new int[]{1, 1, 1, 1, 0, 0, 1}; break;
            case 4: seg = new int[]{0, 1, 1, 0, 0, 1, 1}; break;
            case 5: seg = new int[]{1, 0, 1, 1, 0, 1, 1}; break;
            case 6: seg = new int[]{1, 0, 1, 1, 1, 1, 1}; break;
            case 7: seg = new int[]{1, 1, 1, 0, 0, 0, 0}; break;
            case 8: seg = new int[]{1, 1, 1, 1, 1, 1, 1}; break;
            case 9: seg = new int[]{1, 1, 1, 1, 0, 1, 1}; break;
            default:
                System.out.println("Invalid BCD digit. Enter a value from 0 to 9.");
                return;
        }

        System.out.println("BCD digit: " + bcd);
        System.out.println("Segment order: a b c d e f g");
        System.out.print("Segment outputs: ");
        for (int value : seg) {
            System.out.print(value + " ");
        }
        System.out.println();
    }
}
```